

Chapter 191

Blackmagic Cloud Project Server

This chapter describes setting up and using the Blackmagic Cloud Project Server, allowing you to share projects and collaborate with other DaVinci Resolve users around the world.

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Cloud Project Libraries

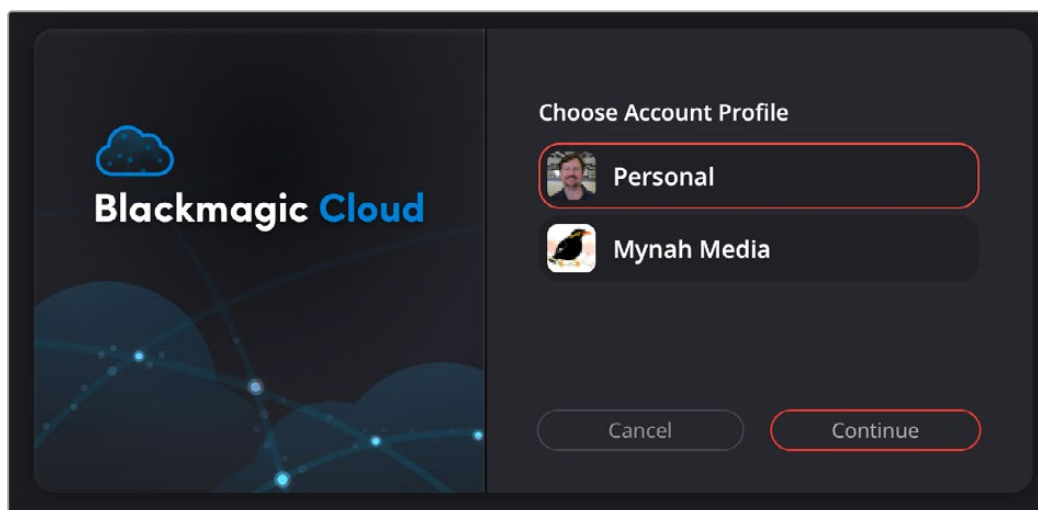
Cloud project libraries are hosted on Blackmagic's Project Library servers on the internet, allowing DaVinci Resolve users to connect and collaborate on the same projects from any location in the world.

Connecting to a Blackmagic Cloud Project Library in DaVinci Resolve

Blackmagic houses project library cloud servers in various locations around the world that users can access for a nominal monthly fee.

To sign into the Blackmagic cloud:

- 1 cloud: Sign up for a Blackmagic ID using your email address at <http://blackmagicdesign.com>
- 2 In the Project Manager window, select Cloud from the Project Library icons in the upper left.
- 3 Chose the Blackmagic Cloud option from the Sign In dialog, and input your Blackmagic ID and password.
- 4 If you are a member of an Organization Profile, you can choose to sign into that account or your personal account.



The Blackmagic Cloud login dialog box

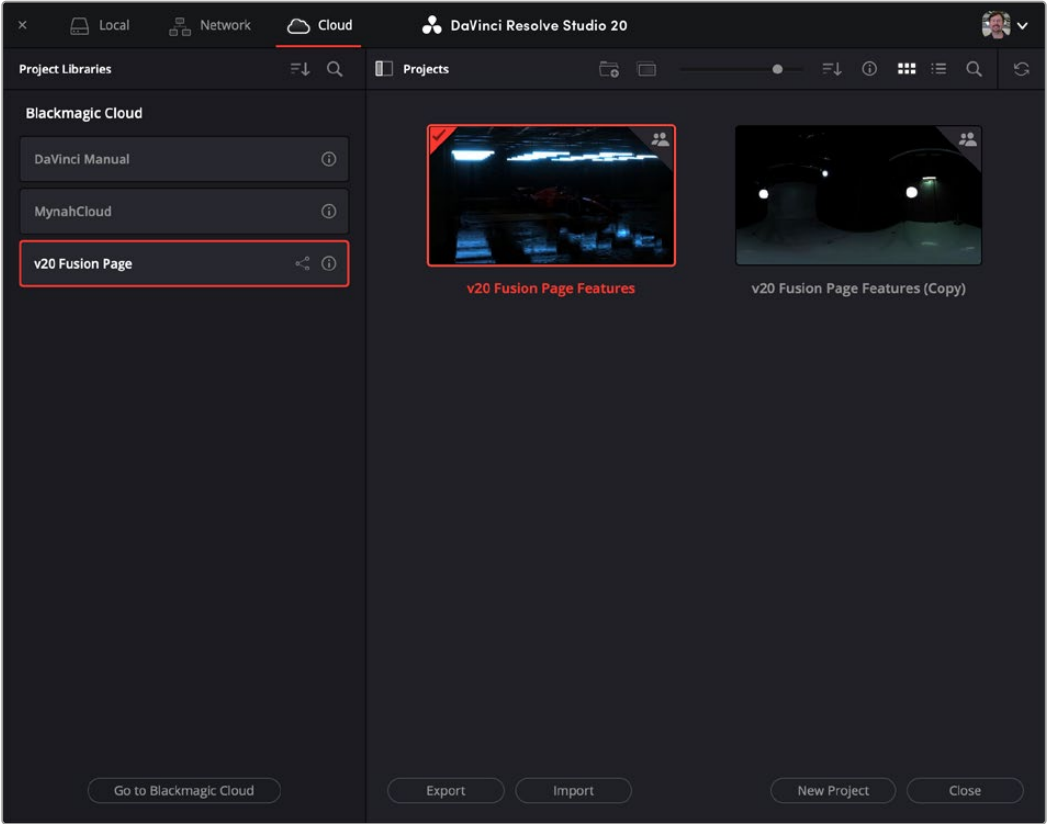
Signing out of the Blackmagic cloud:

- 1 In the Project Manager window, select Cloud from the Project Library icons in the upper left, and then click your account icon in the upper right, and select Log Out.
- 2 Select File > Blackmagic Cloud Account > Log Out.
- 3 Select Preferences > Internet Accounts Blackmagic Cloud > Log Out.

Accessing the Cloud Project Library in DaVinci Resolve

Once connected, cloud project libraries are accessed by clicking on the Show/Hide Project Libraries icon in the upper left of the Project Manager. A sidebar then opens up showing all your connected project libraries. Click on the Cloud icon to open up the cloud project library. It is split into two sections: "My Cloud," which manages all the project libraries that you create, and "Shared with me," which shows project libraries that other users have created but given you shared access to.

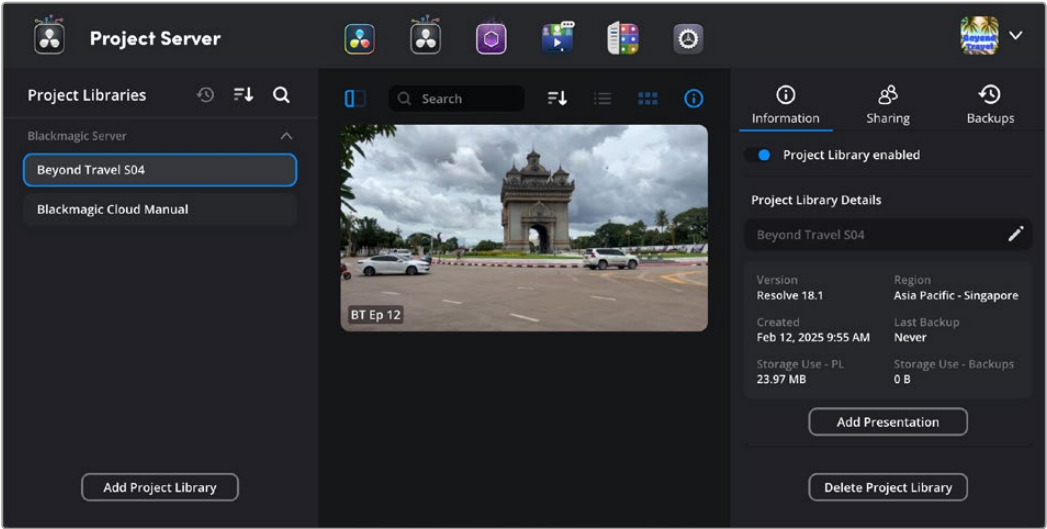
If you don't see your project library immediately, you can click on the circular arrow refresh icon in the upper right of the Project Manager to force a refresh from the servers.



The Blackmagic cloud library

Blackmagic Cloud Project Server on the Web

Cloud project libraries are best managed from within DaVinci Resolve, however there is a web interface as well. This interface lets you see what projects are in each library, search and sort, and create new cloud project libraries.



The Blackmagic Cloud Project Server web interface

When initially opened, the Project Server will show all the active projects in your last used project library. All of your cloud project libraries will be listed in the left hand pane, and you can add new ones by clicking on the Add Project Library button at the bottom.

When a cloud project library is selected, all of the projects it contains will be displayed and can be searched for, sorted, and listed using the respective controls on the right side of the toolbar. As of this writing, on the web interface, projects can only be viewable. In order to create, delete, copy or otherwise manage an individual project, you must do it from within the DaVinci Resolve Project Manager instead

By clicking on the information icon, on the right hand side of the toolbar, you can open the libraries' sharing settings. Here you can add, enable, or disable the project library, or add a new presentation. The Sharing tab shows you who currently has access to the project library, and you can add more users by clicking on the Share Project Library button below. The Backups tab lets you create and restore backups of your project library to the Blackmagic Cloud; you can optionally set automatic daily backups for added safety.

Optimization and Performance of a Cloud Project Library

Before learning about how to create and manage cloud project libraries, it's worthwhile addressing server lag and optimizations. The project library is a database of all the edits, clip metadata, visual effects, color corrections, and audio engineering applied to your timeline. This project library is queried and updated constantly as you use DaVinci Resolve. When the project library is local to, or on the same network as your workstation, these updates happen more or less instantaneously. However, when the project server is half-way around the world on the internet, the speed of light and internet routing start to insert perceptible lag time.

Luckily the majority of the changes required to mitigate this have been done by the DaVinci Resolve team. They have re-written the underlying project library code over a period of several months to optimize it for internet performance, and in most cases the responsiveness will be indistinguishable from using a local project library. However, these types of processes involve intensive and persistent project library operations, and some lag will become apparent when:

- Changing cloud project libraries
- Loading a project from the cloud project library
- Backing up and restoring cloud project libraries

It's important to keep in mind, that once loaded, actually working in DaVinci Resolve will still be as fluid and responsive as you are used to working with local libraries.

TIP: The major optimization that the user can make in cloud library performance is to decide where to physically locate the server. The best performance will be found closest to the city where the team is working. In some cases, the team will be working in multiple cities and countries around the world. In those cases, the server should be hosted in the region of the person who uses the project library the most, and that person is the editor. In the Edit and Cut pages, the project library is written to whenever the user releases the mouse button, while on the Color page the project library is only written to when you select a new clip. So for the best performance, you want to host the server location as close to the editor as possible.

Creating a New Cloud Project Library

Creating a new cloud library requires manually adding it on the Blackmagic Cloud website.

To create a new cloud project library:

- 1 Tap on the Cloud icon to Sign into the Blackmagic cloud server in the Project Manager.
- 2 Tap on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Tap on the “Go to Blackmagic Cloud” button at the bottom of the sidebar, which takes you to the Web Project Server.
- 4 A web browser will open up, automatically taking you to the Blackmagic Cloud web interface.
- 5 Tap on the “Add Project Library” button at the bottom of the sidebar in your web browser.
- 6 Enter a new name for your cloud project library, and then select the region in the world in which you want to host the server. It is best to select a server closest to the project’s editor, and then the Version of Davinci Resolve you want the library to be compatible with.
- 7 Press the Add button.

You can now create or import new projects directly into your new cloud project library.

Deleting or Renaming a Cloud Project Library

If you are finished with a particular cloud project library and want to delete it or wish to change its name you can do so through the Project Libraries interface.

To delete a cloud project library:

- 1 Sign into the Blackmagic cloud server in the Project Manager.
- 2 Click on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Right-click on the library you want to delete in the Project Libraries sidebar, and select Delete from the drop-down menu.
- 4 Click the Delete button on the confirmation dialog box that opens.

Deleting a cloud project library is a permanent and not undoable action. Make sure you have everything you need from this library before you click delete. Once it’s gone, it’s gone.

To rename a cloud project library:

- 1 Sign into the Blackmagic cloud server in the Project Manager.
- 2 Click on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Right-click on the library you want to rename in the Project Libraries sidebar, and enter the new name in the dialog box.
- 4 Click the OK button.

TIP: You can not delete or change the name of the currently connected project library (indicated by an orange highlight around it). In order to do so, you must select and connect to another project library first, and then apply the steps above.

Upgrading a Cloud Project Library

From time to time, new versions of DaVinci Resolve require changes to the way projects are created, which requires project libraries created with older versions of DaVinci Resolve to be upgraded before you can access the projects within. Fortunately, this is a simple process, but it can only be performed in DaVinci Resolve, not the web interface.

To upgrade a cloud project library:

- 1 Sign into the Blackmagic cloud server in the Project Manager.
- 2 Click on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Right-click on the library you want to upgrade in the Project Libraries sidebar, and select Upgrade from the drop-down menu.
- 4 Click the upgrade button on the confirmation dialog box that opens.

Sharing a Cloud Project Library

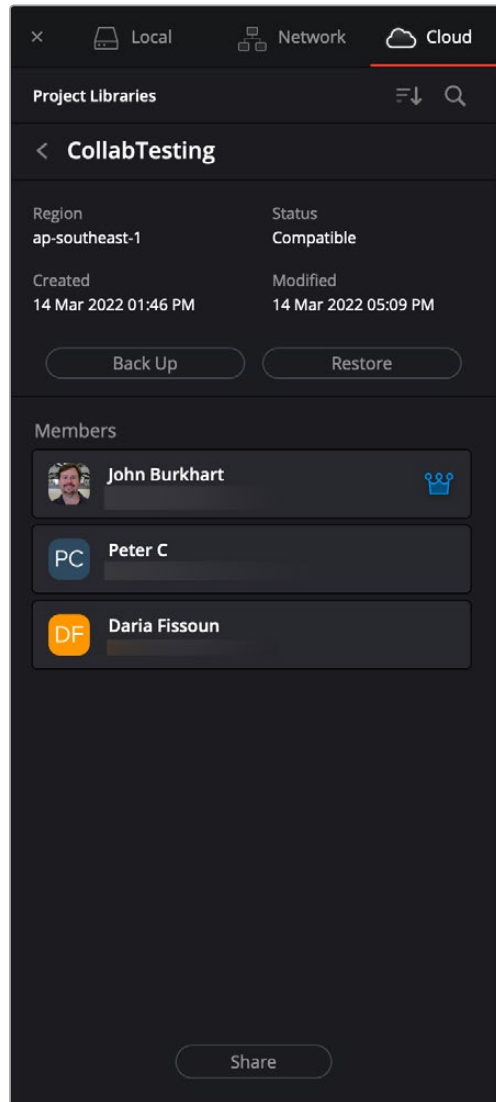
You can share a cloud project library with other users around the world with a Blackmagic ID.

To share a cloud project library:

- 1 Sign into the Blackmagic cloud server in the Project Manager.
- 2 Click on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Click on the Details icon (the circled letter “i”) on the cloud project library you wish to share to open the details settings.
- 4 A members list (or a sharing tab in the web interface) will appear, and your user name and email will be first on the list with a little crown icon, showing that you are the owner of this project library.
- 5 Click the Share button at the bottom of the sidebar.
- 6 Enter the Blackmagic ID (email address) of the person you want to share this project library with.
- 7 Press the Share button.

The user will instantly have access to this shared library, and an email will inform them of their access as well.

IMPORTANT: The users you share your project library with have access to modifying and deleting any projects within that shared library, so be judicious about who you give access to.



The Members section of the Blackmagic cloud library

Removing a User from a Shared Cloud Project Library

If you are the owner of a shared cloud project library, you can remove another shared user's access to it.

Removing a shared user from a cloud project library:

- 1 Sign into the Blackmagic cloud server in the Project Manager.
- 2 Click on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Click on the Details icon (the circled letter "i") on the cloud project library you wish to share to open the details settings.
- 4 A members list (or a sharing tab in the web interface) will appear, showing all users that have access to this project library.
- 5 Right-click on the user you wish to remove and select Remove Member from the drop-down list.

Backing up and Restoring a Cloud Project Library

You can back up and restore a cloud project library in the Blackmagic cloud itself.

To back up a cloud project library:

- 1 Sign into the Blackmagic cloud server in the Project Manager.
- 2 Click on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Click on the Details icon (the circled letter “i”) on the cloud project library you wish to back up to open the details settings.
- 4 Click the Back Up button.
- 5 After some time, a dialog box will appear confirming the backup has been made to the cloud.

To restore an older version of a cloud project library:

- 1 Sign into the Blackmagic cloud server in the Project Manager.
- 2 Click on the Show/Hide Project Libraries icon in the upper left of the Project Manager to expose the sidebar.
- 3 Click on the Details icon (the circled letter “i”) on the cloud project library you wish to restore to an earlier version to open the details settings.
- 4 Click the Restore button.
- 5 Navigate to the version that you want to restore in the Backups list.
- 6 Assign a new name to the restored library.
- 7 Click the Restore button.

Blackmagic Cloud Support for Importing ATEM Projects

If you're working with ATEM switchers, and use the workflow of creating multiple ISO feeds and the accompanying DaVinci Resolve project, you can now log into a Blackmagic Cloud project library, select an existing DaVinci Resolve cloud project, and use the Camera Uploads mechanism to synchronize recorded media to DaVinci Resolve. DaVinci Resolve collaborators can see the edit decisions made from the switcher as a new timeline in that project.

- The DaVinci Resolve project needs to be set to synchronize camera originals to see the uploaded media from the switcher.
- New timelines may use custom settings if the ATEM resolution or frame rate differs from project settings.

Setting up a Cloud-Based Collaboration Workflow

As internet bandwidth has increased over the years, it has recently become possible to collaborate on a project completely online. While once you needed to be in the same building connected to a fast LAN, and Network Attached Storage (NAS), it is now possible to collaborate in real time from all around the world using the internet and cloud storage instead. Below are instructions for setting up a completely cloud-based workflow using the tools and settings in DaVinci Resolve 18 or higher.

This sample workflow consists of Editor A, Colorist B, and Audio Engineer C, all in different parts of the world, and wanting to collaborate on the same project at the same time. The ideal is to minimize the amount of media management involved and to not have to send individual project files back and forth.

Set up the Blackmagic Cloud and a Cloud Library

- Users A, B, and C sign up for the Blackmagic cloud service.
- Users A, B, and C log into the cloud in the Project Manager in their own copies of DaVinci Resolve.
- User A creates a new cloud library, and invites Users B and C to share it. Since User A is the editor, they will locate the cloud library's server nearest themselves.

Set up the Cloud Storage

- Users A, B, and C sign up for a cloud storage provider (Dropbox, iCloud, OneDrive, Google Drive, etc.)
- Users A, B, and C configure their cloud storage so they all can share access to the same cloud-based folder. This folder should be at the top level of the cloud storage. They decide to name the folder Episode 12.
- Users A, B, and C create a file hierarchy system in their shared Episode 12 folder, such as new subfolders for Audio, Proxies, and Graphics.
- Users A, B, and C mount their shared storage folder on their own computers. Editor A adds some logos and still photos to the Graphics folder. Audio Engineer C adds some music and sound effects to the Audio folder. This takes a while to upload and distribute from the cloud, but eventually all users have the same media locally on their computers.

Create the Proxy Media

- Colorist B has the RAW camera masters on a hard drive connected to their system. Since only they need access to the RAW camera files for color grading, they will make low bandwidth proxies for the Editor and Audio Engineer to work with. These files are small enough to upload and store in their cloud storage folder.
- Colorist B creates proxy files of the RAW media in the Blackmagic Proxy Generator application.

For more information on using the Blackmagic Proxy Generator, see *Chapter 8, "Improving Performance, Proxies, and the Render Cache."*

- Colorist B uploads the proxy files to the Proxies folder in their cloud storage.

Setup the DaVinci Resolve Project and Settings

- Colorist B creates the new project in the cloud library, and sets up its resolution and frame rate, etc.
- Colorist B turns on the File > Multiple User Collaboration setting.
- Users A, B, and C open the project and set their individual file paths to their cloud storage folder "Episode 12" in the Path Mapping section of the Project Settings.

For more information on Path Mapping, see *Chapter 6, "Project Settings."*

- Colorist B imports the RAW media from their local hard drive into their Media Pool, and links them to the proxies he uploaded to the Episode12/Proxies folder. At this point, Editor A, and Audio Engineer C now have access to the proxy media, while Colorist B can switch back and forth between RAW media and Proxies as needed. If necessary, Colorist B can also decide to upload the RAW media to the shared folder if space and time allow. While the other users are waiting for this media to upload, if they have the “Prefer Camera Originals” setting checked in Playback > Proxy Handling menu, they can continue to edit using the proxies, and as the RAW media files upload, they will automatically replace the proxy files as they come in.
- Editor A imports their still photos to the Media Pool from the Episode 12/Graphics folder. They immediately become available to Users B and C without relinking.
- Audio Engineer C imports their music tracks to the Media Pool from the Episode 12/Audio folder. They immediately become available to Users A and B without relinking.
- As the users continue to add more media into the shared folder and bring it into the Media Pool, there may be a lag as the media is uploaded to the cloud storage, and then downloaded to the other users. During this time the clip will appear as media offline but will relink automatically once the file finishes its download to the local computer.

Continue working in DaVinci Resolve’s Collaborative Workflow

From here, the editor edits, the colorist colors, and the audio engineer handles the sound design all using DaVinci Resolve’s existing collaborative workflow tools.

For more information on using these tools, see *Chapter 197, “Collaborative Workflow.”*